

Problem Set for 19-May and 22-May

Part I: the Kronecker product, tensor products for matrices

Example Take

• $A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}, B = \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix}, X = \begin{pmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \end{pmatrix}, Y = \begin{pmatrix} y_{11} & y_{12} \\ y_{21} & y_{22} \end{pmatrix}$. Now

1. The matrix equation $AX = Y$ coincides the linear system

$$\begin{pmatrix} a_{11} & 0 & a_{12} & 0 \\ 0 & a_{11} & 0 & a_{12} \\ a_{21} & 0 & a_{22} & 0 \\ 0 & a_{21} & 0 & a_{22} \end{pmatrix} \cdot \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{pmatrix}.$$

2. The matrix equation $XB = Y$ coincides linear system

$$\begin{pmatrix} b_{11} & b_{21} & 0 & 0 \\ b_{12} & b_{22} & 0 & 0 \\ 0 & 0 & b_{11} & b_{21} \\ 0 & 0 & b_{12} & b_{22} \end{pmatrix} \cdot \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{pmatrix}.$$

3. The matrix equation $AXB = Y$ coincides linear system

$$\begin{pmatrix} a_{11}b_{11} & a_{11}b_{21} & a_{12}b_{11} & a_{12}b_{21} \\ a_{11}b_{12} & a_{11}b_{22} & a_{12}b_{12} & a_{12}b_{22} \\ a_{21}b_{11} & a_{21}b_{21} & a_{22}b_{11} & a_{22}b_{21} \\ a_{21}b_{12} & a_{21}b_{22} & a_{22}b_{12} & a_{22}b_{22} \end{pmatrix} \cdot \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{pmatrix}.$$

where we write

$$\begin{pmatrix} a_{11}b_{11} & a_{11}b_{21} & a_{12}b_{11} & a_{12}b_{21} \\ a_{11}b_{12} & a_{11}b_{22} & a_{12}b_{12} & a_{12}b_{22} \\ a_{21}b_{11} & a_{21}b_{21} & a_{22}b_{11} & a_{22}b_{21} \\ a_{21}b_{12} & a_{21}b_{22} & a_{22}b_{12} & a_{22}b_{22} \end{pmatrix} = \begin{pmatrix} a_{11}B^T & a_{12}B^T \\ a_{21}B^T & a_{22}B^T \end{pmatrix} =: A \otimes B^T.$$

Definition (Kronecker product of matrices) For $P \in K^{k \times l}, Q \in K^{m \times n}$, define

$$P \otimes Q = \begin{pmatrix} p_{11}Q & p_{12}Q & \cdots & p_{1l}Q \\ p_{21}Q & p_{22}Q & \cdots & p_{2l}Q \\ \vdots & \vdots & \ddots & \vdots \\ p_{k1}Q & p_{k2}Q & \cdots & p_{kl}Q \end{pmatrix}.$$

Clearly, $P \otimes Q \in K^{km \times ln}$.

Exercise 1 Verify the following statements yourself (**not a homework**):

1. $\otimes$ is an binary operation which is associative;
2. $\otimes$ is bilinear on both sides;
3. $(A \otimes B) \cdot (C \otimes D) = (A \cdot C) \otimes (B \cdot D)$;
4. $(A \otimes B)^T = A^T \otimes B^T$;
5. $\text{rank}(A \otimes B) = \text{rank}(A) \cdot \text{rank}(B)$;

Now we assume that A and B are square matrices:

1. $(A \otimes B)^{-1} = A^{-1} \otimes B^{-1}$;
2. $\det(A \otimes B) = (\det A)^{\dim A} (\det B)^{\dim B}$;
3. $\text{tr}(A \otimes B) = \text{tr}(A) \cdot \text{tr}(B)$;
4. Let (λ, v) and (μ, u) be eigenpairs of A and B , respectively. Then $A \otimes B$ has $(\lambda\mu, v \otimes u)$ as an eigenpair;

Exercise 2 Let J_A and J_B denote the Jordan form of A and B respectively (all matrices are in $\mathbb{F}^{n \times n}$).

1. Write down the Jordan form of the linear transformation $X \mapsto AXA^T$;
2. Write down the Jordan form of the linear transformation $X \mapsto AXA$;
3. Write down the Jordan form of the linear transformation $X \mapsto AX - XA$;
4. Write down the Jordan form of the linear transformation $X \mapsto AX - XA^T$;
5. Write down the Jordan form of the linear transformation $X \mapsto AXB$.

Tip

We know that $A \sim A^T$ over arbitrary field, since $(\lambda I - A)$ and $(\lambda I - A^T)$ are equivalent.

Exercise 3 Show that $\dim \ker[X \mapsto (AX - XA^T)] \geq n$, and explain when the equality holds.

Exercise 4 Show that there exists $\{0, 1\}$ -matrices A and B such that $A \otimes B$ and $B \otimes A$ are not similar.

Tip

For square matrices A and B , it is clear that $A \otimes B \sim B \otimes A$. Proof of the statement: By linear isomorphism of $(-)^T$, $[X \mapsto AXB^T]$ and $[Y \mapsto BYA^T]$ stand for the same linear map under different basis. Hence $(A \otimes B) \sim (B^T \otimes A^T)$. Since $A \sim A^T$ and $B \sim B^T$, one has $(A \otimes B) \sim (B \otimes A)$.

Exercise (optional) If you know the adjacency matrices $A(G_i)$ for a simple graph G_i , then discover the graph operations $\boxplus$ and $\boxtimes$ such that

1. $A(G_1) \otimes A(G_2) = A(G_1 \boxtimes G_2)$;
2. $A(G_1) \otimes I + I \otimes A(G_2) = A(G_1 \boxplus G_2)$.

 Tip

Kronecker 积 $\otimes$ 的本质是指标的运算, 图论语言可以将这一指标运算可视化.

Part II: the tensor product in general, and the use of 1 : 1 correspondence

 Important

Throughout, $\sum$ is always a finite sum.

Definition (Tensor product of two finite dimensional vector spaces). Let $\{u_i\}_{1 \leq i \leq n}$ be a basis of U , and $\{v_j\}_{1 \leq j \leq m}$ be a basis of V . Now one can show the 1 : 1 correspondence of the following three types of functions.

1. a bilinear map from U, V to $\mathbb{F}$;
2. a map of the type $f : S \rightarrow \mathbb{F}$ with property P, where
 - $S := \{M \in \mathbb{F}^{n \times m} \mid \text{rank}(M) = 1\}$, the subspace of rank one matrices;
 - P means that, once there is an equality of linear combination in S (e.g., $\sum c_i s_i = 0$ for $s_i \in S$), one has $\sum c_i f(s_i) = 0$.
3. a linear map $\text{Hom}_{\mathbb{F}}(\mathbb{F}^{n \times m}, \mathbb{F})$.

We identify $U \otimes V := \mathbb{F}^{n \times m}$ in 3. by chasing the basis $\{u_i\}_{1 \leq i \leq n}$ and $\{v_j\}_{1 \leq j \leq m}$ from 1. to 3.:

- $U \otimes V$ has a basis $u_i \otimes v_j$, i.e., the $E_{i,j}$ matrices in the space $\mathbb{F}^{n \times m}$;
- An element of $U \otimes V$ is of the form $\sum \lambda_i (u_i \otimes v_i)$, i.e., the sum of $E_{i,j}$'s;
- Say a $x \in U \otimes V$ is an simple tensor, whenever there is $u \in U$ and $v \in V$ such that $x = u \otimes v$, i.e., the rank-one matrix in $\mathbb{F}^{n \times m}$;
- The tensor-rank of $x \in U \otimes V$ is the least number of simple tensors summing up to x , i.e., 使用相抵标准型将矩阵写作秩 1 矩阵的和.

Definition (The definition of tensor product in general). We consider the properties that a hypothetical object $U \otimes V$ would possess:

1. an object of $x \in U \otimes V$ is the finite sum of simple tensors, e.g., $x = \sum u_t \otimes v_t$ for some $u_t \in U$ and $v_t \in V$;
2. there is a canonical bilinear map $\theta_{U,V} : U \otimes V \rightarrow U \otimes V$ sending the pair (u, v) to a simple tensor $u \otimes v$, which means that

1. $(u + u') \otimes v$ and $u \otimes v + u' \otimes v$ are the same elements,
 2. $u \otimes (v + v')$ and $u \otimes v + u \otimes v'$ are the same elements,
 3. $\lambda(u \otimes v)$, $(\lambda u) \otimes v$ and $u \otimes (\lambda v)$ are the same elements;
3. any bilinear form $U \& V \rightarrow W$ corresponds to a linear map from $U \otimes V$ to W , which means that
1. once we have a bilinear map $B : U \& V \rightarrow W$, we have a linear map $U \otimes V \rightarrow W$ determined by the image of simple tensors:

$$u \otimes v \mapsto B(u, v).$$

2. once we have a linear map $\varphi : U \otimes V \rightarrow W$, there is a bilinear map $\varphi \circ \theta_{U,V}$.

③ Note

In short, the linear space $\text{Hom}_{\mathbb{F}}(U, \text{Hom}_{\mathbb{F}}(V, W))$ (not $\text{Hom}_{\mathbb{F}}(U \times V, W)$) stands for the set of bilinear maps from $U \& V$ to W . The 1 : 1 correspondence means the bijection

$$\text{Hom}_{\mathbb{F}}(U, \text{Hom}_{\mathbb{F}}(V, W)) \rightarrow \text{Hom}_{\mathbb{F}}(U \otimes V, W), \quad B \mapsto B \circ \theta_{U,V}$$

Such an bijective is also linear (verify it your self). Hence, the above is an linear isomorphism.

The adjunction: $\text{Hom}(A, \textcolor{red}{R}B) \simeq \text{Hom}(\textcolor{blue}{L}A, B)$.

Definition ($f \otimes f'$). For linear maps $f : U \rightarrow V$ and $f' : U' \rightarrow V'$, the bilinear map

$$U \& U' \rightarrow V \& V' \rightarrow V \otimes V' \quad (u, u') \mapsto f(u) \otimes f'(u')$$

corresponds to a unique linear map $U \otimes U' \rightarrow V \otimes V'$ sending simple tensor $u \otimes u'$ to $f(u) \otimes f'(u')$. We denote

$$f \otimes f' : U \otimes U' \rightarrow V \otimes V', \quad \sum u \otimes u' \mapsto \sum f(u) \otimes f'(u').$$

Example (The magic of 1 : 1 correspondence). We show that $(g \otimes g') \circ (f \otimes f') = (g \circ f) \otimes (g' \circ f')$ are the same linear maps from $U \otimes U'$ to $W \otimes W'$, where

$$U \xrightarrow{f} V \xrightarrow{g} W, \quad U' \xrightarrow{f'} V' \xrightarrow{g'} W'.$$

Step 1 $(g \otimes g') \circ (f \otimes f') = (g \circ f) \otimes (g' \circ f')$ whenever $(g \otimes g') \circ (f \otimes f') \circ \theta_{U,U'} = (g \circ f) \otimes (g' \circ f') \circ \theta_{U,U'}$ are the same linear map, where $\theta_{U,U'} : U \& U' \rightarrow U \otimes U'$ is the canonical bilinear map.

Step 2 We consider $(g \otimes g') \circ (f \otimes f') \circ \theta_{U,U'}$. By definition of $f \otimes f'$, we have $(f \otimes f') \circ \theta_{U,U'} = \theta_{V,V'} \circ (f \& f')$, the same bilinear map from $U \& U'$ to $V \otimes V'$. Hence

$$(g \otimes g') \circ (f \otimes f') \circ \theta_{U,U'} = (g \otimes g') \circ \theta_{V,V'} \circ (f \& f') = \theta_{W,W'} \circ (g \& g') \circ (f \& f').$$

Step 2' We consider $(g \circ f) \otimes (g' \circ f') \circ \theta_{U,U'}$. By definition, one has

$$(g \circ f) \otimes (g' \circ f') \circ \theta_{U,U'} = \theta_{W,W'} \circ ((g \circ f) \& (g' \circ f')).$$

Step 3 It is clear that $((g \circ f) \& (g' \circ f'))$ and $(g \& g') \circ (f \& f')$ are the same thing from $U \& U' \rightarrow W \& W'$. This complete the proof.

Exercise 5 Show by 1 : 1 correspondence that, there exists a unique isomorphism

$$\Phi_{U,V,W} : U \otimes (V \otimes W) \rightarrow (U \otimes V) \otimes W,$$

sending simple tensor $u \otimes (v \otimes w)$ to $(u \otimes v) \otimes w$.

(optional) Moreover, such Φ is natural, which means that for $f : U_f \rightarrow V_f$, $g : U_g \rightarrow V_g$ and $h : U_h \rightarrow V_h$, one has

$$((f \otimes g) \otimes h) \circ \Phi_{U_f, U_g, U_h} = \Phi_{V_f, V_g, V_h} \circ (f \otimes (g \otimes h))$$

Exercise 6 Show that for arbitrary linear surjection p , the linear map $\text{id}_U \otimes p$ is also a surjection.

Exercise 7 Show that for arbitrary linear injection i , the linear map $\text{id}_U \otimes i$ is also an injection.

💡 Tip

You are allowed to use the following axiom: For any vector subspace $S \subseteq L$, any linear map $f : S \rightarrow W$ extends to $\tilde{f} : L \rightarrow W$.